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United States Patent Application Publication

20250268606

Kind Code

A1

Publication Date

August 28, 2025

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Medical Clamping Device

Abstract

A clamping device includes a flexible first part that has a first side wall and a second side wall that is spaced opposite from and is joined to the first side wall. The first side wall and the second side wall are flexible and being configured to move toward one another. The first part includes a locking bar coupled to the first side wall and the second side wall includes a locking protrusion that is configured to engage and lock with the locking bar to close off and lock the first part. The first part has an open unlocked position, in which the locking bar is disengaged with the locking protrusion, for receiving a traumatized digit. The device includes a cushioned second part that lines an inner face of the first part. The cushioned second part includes rounded inner portions configured to support a digit when the device is in a closed locked position for capturing the traumatized digit between the cushioned second part.

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Family ID: 1000007728213

Appl. No.: 18/599694

Filed: March 08, 2024

Related U.S. Application Data

us-provisional-application US 63558330 20240227

Publication Classification

Int. Cl.: A61B17/122 (20060101); A61B17/12 (20060101)

Background/Summary

CROSS REFERENCE TO RELATED APPLICATION [0001] The present application claims priority to and the benefit of U.S. patent application Ser. No. 63/558,330, filed Feb. 27, 2024, which is hereby expressly incorporated by reference in its entirety.

TECHNICAL FIELD

[0002] The present disclosure relates to medical devices and more specifically, to a medical clamping device that stops bleeding of a patient's digit (e.g., a finger) in situations where the digit has undergone trauma and is bleeding.

BACKGROUND

[0003] The fingers and toes are richly vascularized and bleed heavily during trauma. Trauma can be in the form of a laceration to the finger or toe or a finger avulsion or similar type of trauma. Tourniquets are often applied to the digits to control and minimize blood loss providing a clear surgical field that allows for delineation of structures, thereby reducing risk for damage to underlying vital structures and increasing the prospects for positive outcomes. As is known in the field, a tourniquet is a constricting device used to control venous and arterial circulation to an extremity for a period of time.

[0004] A field obscured by actively oozing blood creates a nearly impossible situation for fully evaluating the presence of joint capsule or tendon injury and foreign body. The “bloodless field” improves precision in repair and reduces risk of inadvertently damaging underlying vital structures such as tendons, etc. There are many types of tourniquets that are intended to provide a bloodless field. One type of product is a ring-like band that constricts blood flow across the entire diameter of the digit.

[0005] The present disclosure describes and illustrates an improved clamping device to treat a digit that has undergone trauma and is bleeding.

SUMMARY

[0006] A clamping device according to one embodiment disclosed herein includes a flexible first part that has a first side wall and a second side wall that is spaced opposite from and is joined to the first side wall. The first side wall and the second side wall are flexible and being configured to move toward one another. The first part includes a locking bar coupled to the first side wall and the second side wall includes a locking protrusion that is configured to engage and lock with the locking bar to close off and lock the first part. The first part has an open unlocked position, in which the locking bar is disengaged with the locking protrusion, for receiving a traumatized digit. The clamping device includes a cushioned second part that lines an inner face of the first part. The cushioned second part includes rounded inner portions configured to support a digit when the device is in a closed locked position for capturing the traumatized digit between the cushioned second part.

[0007] In one embodiment, a clamping device includes a first part that has a first side wall and a second side wall that is spaced opposite from and is joined to the first side wall. The first side wall and the second side wall are flexible and are configured to move toward one another, wherein the first part includes a locking bar and the second side wall includes a locking protrusion that is configured to engage and lock with the locking bar. The clamping device further includes a cushioned second part that lines an inner face of the first part. The clamping device has an open

unlocked position for receiving a traumatized digit and a closed locked position for capturing the traumatized digit between the cushioned second part.

Description

BRIEF DESCRIPTION OF THE DRAWING FIGURES

[0008] FIG. **1** is a front perspective view of the clamping device in a first position;
[0009] FIG. **2** is a front perspective view of the clamping device in a second position;
[0010] FIG. **3** is a front perspective view of the clamping device in a third position;
[0011] FIG. **4** is a front perspective view of the clamping device in a fourth position;
[0012] FIG. **5** is a front perspective view of the clamping device in a fifth position;
[0013] FIG. **6** is a perspective showing an injured finger;
[0014] FIG. **7A** a perspective view of the clamping device with the injured finger being clamped;
[0015] FIG. **7B** is a side view thereof;
[0016] FIG. **8** is a perspective view of the clamping device with the injured finger being clamped and a delivery needle;
[0017] FIG. **9A** a perspective view of the clamping device with the injured finger being clamped; and
[0018] FIG. **9B** is a side view thereof.

DETAILED DESCRIPTION OF CERTAIN EMBODIMENTS

[0019] FIGS. **1-9B** illustrate a clamping device **100** for use with a digit, such as a finger **10**. The clamping device **100** is configured to stop the bleeding of a patient's digit (finger) in situations where the digit has undergone trauma and is bleeding. FIGS. **1-5** show the structure and movements of the clamping device **100**, while FIGS. **6-9B** show a method of treatment using the clamping device **100**.

[0020] The clamping device **100** is generally formed of a first part **200** and a second part **300**. The first part **200** can be generally thought of as being an outer rigid element and the second part **300** can be generally thought of as being an inner cushioning element. The first part **200** is configured to maintain shape and tension and the second part **300** is a cushioning element that makes contact with the relevant digit.

First Part **200**

[0021] As described herein, the first part **200** moves between an open (unlocked) position and a closed (locked) position. The first part **200** has a closed first end **201** and an open second end **203**.
[0022] The first part **200** includes a first side wall **210** and a second side wall **220** that is joined, in a continuous manner, to the first side wall **210** along a top wall section **230**. Between the first side wall **210** and the second side wall **220** there is an open space **215** for receiving the digit. The combined first side wall **210**, second side wall **220** and the top wall section **230** generally have a U shape. While the first part **200** is rigid, it also has a degree of flexing as shown in the figures. In particular, the first and second side walls **210**, **220** can be considered to be fingers or prongs that are free to flex when a force is applied thereof, whereby, as shown, the first and second side walls **210**, **220** can move toward one another or away from one another. For example, the first part **200** can be a molded plastic piece.

[0023] Along an inner face of the top wall section **230**, there is a coupling channel **235**. The coupling channel **235** can be at the apex of the first part **200**.

[0024] At or proximate the first end **201**, the first part **200** includes a pair of outer levers **240** that protrude and extend radially outward. For example, one outer lever **240** is integral to and extends outward from the first side wall **210** and the other outer lever **240** is integral to and extends outward from the second side wall **220**. As shown, the pair of outer levers **240** can have curved shapes, including concave sections, to accommodate the thumb or finger of the user and allow a

pressing action. The outer levers **240** are rigid but have some degree of flexing. As shown in FIGS. **1** and **2**, the outer levers **240** are configured such that when an inward force is applied thereto, it results in opposite motion in the first and second side walls **210**, **220** in that the first and second side walls **210**, **220** move away from one another. In other words, the first and second side walls **210**, **220** open up and the open space **215** between the first and second side walls **210**, **220** increases in area.

[0025] The outer levers **240** can have a contoured outer surface, such as ribbing, to increase grip.

[0026] A bottom end of each of the first and second side walls **210**, **220** can include a flange **250** that protrudes outward. The flange **250** can be a rounded flange that is formed so as to define an inner channel **255** that faces inward and is in communication with the open space **215**. As described herein, this inner channel **255** is configured to capture the second part **300**. The channel **255** can have rounded inner walls. The two flanges **250** and two inner channels **255** are opposite one another.

[0027] The bottom end of the second side wall **220** has a locking protrusion **275** that can act as a locking pawl or tooth. The locking protrusion **275** extends downward and is angled outward. The locking protrusion **275** is preferably an integral part of the first part **200**.

[0028] The first part **200** also includes an elongated toothed locking bar (toothed flange) **260** that is coupled to the bottom of the first side wall **210**. The toothed locking bar **260** has a fulcrum **270** about which the locking bar **260** pivots and more specifically, moves between an open position and a closed position. The fulcrum **270** thus represents the connection location between the locking bar **260** and the first side wall **210**.

[0029] The length of the locking bar **260** is such that the free end thereof extends laterally beyond the flange **250** associated with the second side wall **220**.

[0030] The locking bar **260** has teeth **265** formed along an upper surface of the locking bar **260**. The teeth **265** are formed at an angle along the upper surface that is complementary to the angle and structure of the locking protrusion **275**. In other words, the locking protrusion **275** contacts and engages the teeth **265** and in particular, is positioned between two teeth **265** resulting in the locking bar **260** being coupled to the second side wall **220**. The locking bar **260** is thus engaged and locked to the second side wall **220**, resulting in the space **215** being closed off. The locking bar **260** can have a pair of downwardly extending legs **269**. The legs **269** can be located at or near the ends of the locking bar **260** and extend laterally outward. The legs **269** allow the device **100** to stand up (be upright) on a flat surface.

[0031] FIGS. **2** and **3** illustrate the locking action and movements of the device **100**. For example, in FIG. **3**, the user is pressing inward the first and second side walls **210**, **220** which causes the locking protrusion **275** to also move inward and the user can direct the locking protrusion **275** into and between teeth **265** formed on the locking bar **260** resulting again in the locking action.

Second Part **300**

[0032] As mentioned, the second part **300** is a cushioning element that makes contact with the relevant digit. The second part **300** has a shape that is complementary to the inside of the first part **200** since the second part **300** lines the inner face of the first part **200**. The second part **300** lines the inner surface of the first part **200**.

[0033] The second part **300** is an elongated structure that has a first end **302** and an opposite second end **304**. The first end **302** has a shape that is received into and captured within the inner channel **255** of the flange **250** that is part of the first side wall **210** and similarly, the second end **304** of the second part **300** is received within and captured within the inner channel **255** of the flange **250** that is part of the second side wall **220**. This secures the two opposite ends of the second part **300** to the first part **200**. To secure the center region of the second part **300** to the first part **200**, a center flange or protrusion or bead **320** of the second part **300** is received into the coupling channel **235**. This now results in the entire second part **300** lining the inner surface of the first part **200** since the second part **300** is secured to the first part **200** at three distinct locations, namely, the two ends and

center. However this friction fit can be reversed and with applied force, the second part **300** can be detached from the first part **200**.

[0034] As shown, the inner face of the second part **300** can be contoured in view of its intended use. For example, as shown, the hollow interior of the second part **300** can have an enlarged upper space **330**, a smaller waist **332** below the enlarged upper space **330** and an outwardly flared lower space **334**. In particular, the second part **300** can include a pair of inwardly directed rounded portions **301** that act to support the contained finger **10** when the device **100** is closed. These rounded portions **301** define the waist **332** and when contained the digit (finger **10**) is located within the enlarged upper space **330** and rests on the waist **332**.

[0035] The construction of the clamping device **100** allows the patient to rest their hand while the traumatized digit remains raised for care.

[0036] The second part **300** is formed of a cushioned material, such as silicone.

Method of Treatment (FIGS. 6-9)

[0037] FIG. 6 generally shows trauma at the end of one finger **10**.

[0038] With additional reference to FIGS. 1-5, in its initial state (FIG. 1), the clamping device **100** is in the unlocked position and its aperture (space **115**) can be adjusted by compressing the two outer levers **240** towards each other as shown in FIG. 2. As mentioned, to lock the device **100** in place, the user squeezes the first and second side walls **210**, **220** and the locking protrusion **275** is brought into engagement with the teeth **265**. As this locking action occurs, the second part **300** contacts and applies pressure to the contained digit (finger **10**) as shown in FIGS. 3, 7A and 7B.

[0039] After the device **100** is secured and the bleeding is controlled, the traumatized area can now be worked on.

[0040] When no longer requires, the device **100** can be unlocked by lowering the locking bar **260**, thereby disengaging the locking protrusion **275** from the teeth **265** and unlocking the device **100**. Once unlocked, the second end of the device **100** is open and the traumatized finger can be removed from the device **100**.

[0041] As shown in FIGS. 9A and 9B, if required, the device **100** may be adjusted to further facilitate care.

[0042] The present device **100** provides a number of advantages over traditional clamping devices. For example, in ring-like devices, oftentimes the clamping device must pass over the damaged area, causing possible extreme discomfort to the patient, and possibly causing additional damage. The device **100** need not make contact with the affected digit in the same manner, as its aperture (space **215**) can be adjusted to accommodate for many injuries to the finger. Therefore, the ease of application reduces risk of additional trauma in its application and removal. The pressure exerted by the device **100** is inherently adjustable due to the array of locking teeth **265**. This is important as the caretaker need only apply the minimum necessary pressure to control bleeding, which as the added benefit of minimizing risk due to blocking blood circulation. In practice, the caretaker may need to check the blood circulation and tissue perfusion while tending to the traumatized digit. Using other tourniquet devices, this would require application of multiple device unites because they are unable to release pressure during the procedure. In contrast, since the device **100** is adjustable, multiple sized devices are not required. Another adjacent is shown in FIGS. 7A and 7B in that, when applied, the device **100** allows for physical support of the patient's fingers and hand to facilitate application of sterile- strips or cauterization. FIG. 8 shows that when the device **100** is applied, local anesthetics can be applied as by using a needle **20**. In both locking the device **100**, as well as releasing the device, a minimum of one hand may be used if operating conditions allow, whereas other such devices require two hands.

[0043] Unlike other tourniquet-like devices, the present clamping device **100** involves controlling bleeding by targeting the main arteries and veins, without blocking peripheral circulation. In traditional products, pressure is applied around the entire digit, whereas, in the present clamping device **100**, pressure is applied specifically to arteries and veins due to the unique configuration of

the clamping device **100**. The present clamping device **100** thus blocks blood flow by compressing arteries, as opposed to entire digit constriction. This results from the construction and shape of the clamping device.

[0044] The present clamping device **100** also allows for a variable range of pressures to facilitate care.

[0045] Various embodiments of systems, devices, and methods have been described herein. These embodiments are given only by way of example and are not intended to limit the scope of the claimed inventions. It should be appreciated, moreover, that the various features of the embodiments that have been described may be combined in various ways to produce numerous additional embodiments. Moreover, while various materials, dimensions, shapes, configurations and locations, etc. have been described for use with disclosed embodiments, others besides those disclosed may be utilized without exceeding the scope of the claimed inventions.

[0046] Persons of ordinary skill in the relevant arts will recognize that the subject matter hereof may comprise fewer features than illustrated in any individual embodiment described above. The embodiments described herein are not meant to be an exhaustive presentation of the ways in which the various features of the subject matter hereof may be combined. Accordingly, the embodiments are not mutually exclusive combinations of features; rather, the various embodiments can comprise a combination of different individual features selected from different individual embodiments, as understood by persons of ordinary skill in the art. Moreover, elements described with respect to one embodiment can be implemented in other embodiments even when not described in such embodiments unless otherwise noted.

[0047] Although a dependent claim may refer in the claims to a specific combination with one or more other claims, other embodiments can also include a combination of the dependent claim with the subject matter of each other dependent claim or a combination of one or more features with other dependent or independent claims. Such combinations are proposed herein unless it is stated that a specific combination is not intended.

[0048] Any incorporation by reference of documents above is limited such that no subject matter is incorporated that is contrary to the explicit disclosure herein. Any incorporation by reference of documents above is further limited such that no claims included in the documents are incorporated by reference herein. Any incorporation by reference of documents above is yet further limited such that any definitions provided in the documents are not incorporated by reference herein unless expressly included herein.

[0049] For purposes of interpreting the claims, it is expressly intended that the provisions of 35 U.S.C. § 112 (f) are not to be invoked unless the specific terms “means for” or “step for” are recited in a claim.

Claims

1. A clamping device comprising: a first part that has a first side wall and a second side wall that is spaced opposite from and is joined to the first side wall, the first side wall and the second side wall being flexible and being configured to move toward one another, wherein the first part includes a locking bar and the second side wall includes a locking protrusion that is configured to engage and lock with the locking bar; and a cushioned second part that lines an inner face of the first part; wherein the clamping device has an open unlocked position for receiving a traumatized digit and a closed locked position for capturing the traumatized digit between the cushioned second part.
2. The clamping device of claim 1, wherein the first part has a pair of outer levers at a top end of the first part that when compressed toward one another are configured to separate the first and second side walls.
3. The clamping device of claim 1, wherein the locking bar is attached to the first side wall at a fulcrum.

- 4.** The clamping device of claim 3, wherein the locking bar comprises an elongated cantilevered structure that has a free end proximate the second side wall.
 - 5.** The clamping device of claim 1, wherein the locking bar has teeth formed along a top surface thereof and the locking protrusion engages the teeth to lock the locking bar to the second side wall.
 - 6.** The clamping device of claim 1, wherein the locking bar has a pair of downwardly extending legs on which the device can stand upright.
 - 7.** The clamping device of claim 1, wherein the second part is attached to the first part at at least three locations.
 - 8.** The clamping device of claim 7, wherein the second part is attached to the first part at a first end, a second end and a center of the second part.
 - 9.** The clamping device of claim 1, wherein the second part is disposed above the locking bar.
 - 10.** The clamping device of claim 1, wherein the second part is contoured and includes an enlarged upper opening, a narrow waist, and a flared lower opening.
 - 11.** The clamping device of claim 1, wherein the first part includes a first flange at a bottom of the first side wall and a second flange at a bottom of the second side wall, the first flange including a first inner channel and the second flange including a second inner channel.
 - 12.** The clamping device of claim 11, wherein the locking protrusion is formed adjacent the second inner channel.
 - 13.** The clamping device of claim 11, wherein the second part is received at a first end within the first inner channel and is received at a second end within the second inner channel.
 - 14.** A clamping device comprising: a flexible first part that has a first side wall and a second side wall that is spaced opposite from and is joined to the first side wall, the first side wall and the second side wall being flexible and being configured to move toward one another, wherein the first part includes a locking bar coupled to the first side wall and the second side wall includes a locking protrusion that is configured to engage and lock with the locking bar to close off and lock the first part, the first part having an open unlocked position, in which the locking bar is disengaged with the locking protrusion, for receiving a traumatized digit; and a cushioned second part that lines an inner face of the first part, the cushioned second part including rounded inner portions configured to support a digit when the device is in a closed locked position for capturing the traumatized digit between the cushioned second part.
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